MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

61. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero vectors and $\hat{n}$ is a unit vector perpendicular to $\vec{c}$ such that $\vec{a} = \alpha\vec{b} - \hat{n}, (\alpha \neq 0)$ and $\vec{b}\cdot\vec{c} = 12$, then $\left|\vec{c}\times\left(\vec{a}\times\vec{b}\right)\right|$ is equal to

(1) 9 (2) 6

(3) 12 (4) 15

Answer (3)

Sol. $\hat{n} = \alpha\vec{b} - \vec{a}$

$$\vec{c}\times\left(\vec{a}\times\vec{b}\right) = \left(\vec{c}\cdot\vec{b}\right)\vec{a} - \left(\vec{c}\cdot\vec{a}\right)\vec{b}$$

$$= 12\vec{a} - \left(\vec{c}\cdot\left(\alpha\vec{b} - \hat{n}\right)\right)\vec{b}$$

$$= 12\vec{a} - (12\alpha - 0)\vec{b}$$

$$= 12\left(\vec{a} - \alpha\vec{b}\right)$$

$$\therefore \quad \left|\vec{c}\times\left(\vec{a}\times\vec{b}\right)\right| = 12$$

62. If the coefficient of x^{15} in the expansion of $\left(ax^3 + \frac{1}{bx^{1/3}}\right)^{15}$ is equal to the coefficient of x^{-15} in the expansion of $\left(ax^{1/3} - \frac{1}{bx^3}\right)^{15}$, where a and b are positive real numbers, then for each such ordered pair (a, b)

(1) $ab = 1$ (2) $a = b$

(3) $a = 3b$ (4) $ab = 3$

Answer (1)

Sol. For $\left(ax^3 + \frac{1}{bx^{\frac{1}{3}}}\right)$

$$T_{r+1} = {}^{15}C_r\left(ax^3\right)^{15-r}\left(\frac{1}{bx^{\frac{1}{3}}}\right)^1$$

$$\therefore \quad \text{For } x^{15} \rightarrow 3(15-r) - \frac{r}{3} = 15$$

$$\Rightarrow \quad 30 = \frac{10r}{3} \Rightarrow r = 9$$

Similarly, for $\left(ax^{\frac{1}{3}} - \frac{1}{bx^3}\right)^{15}$

$$T_{r+1} = {}^{15}C_r\left(ax^{\frac{1}{3}}\right)^{15-r}\left(-\frac{1}{bx^3}\right)^2$$

$$\therefore \quad \text{For } x^{-15} \rightarrow \frac{15-r}{3} - 3r = -15 \Rightarrow r = 6$$

$$\therefore \quad {}^{15}C_9\frac{a^6}{b^9} = {}^{15}C_6\frac{a^9}{b^6} \Rightarrow ab = 1$$

63. The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is

(1) 4 (2) 3

(3) 2 (4) 0

Answer (1)

Sol. $y' = 270x^4 - 540x^3 - 210x^2 + 360x + 210$

Slope of normal $= -\frac{1}{90}$

$\therefore$ Slope of tangent = 90

$\therefore$ Number of normal will be number of solutions of

$270x^4 - 540x^3 - 210x^2 + 360x + 210 = 90$

$\Rightarrow$ $9x^4 - 18x^3 - 7x^2 + 12x + 4 = 0$

$\therefore$ $x = 1, 2, -\frac{1}{3}, -\frac{2}{3}$ are roots

64. The line l_1 passes through the point (2, 6, 2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is

(1) $\frac{19}{3}$ (2) $\frac{13}{3}$

(3) 9 (4) 7

Answer (3)

Sol. Equation of $l_1 = \frac{x-2}{2} = \frac{y-6}{1} = \frac{z-2}{-2}$

Shortest distance with $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is

$$\text{S.d} = \left|\frac{\begin{vmatrix} 3 & 10 & 2 \\ 2 & 1 & -2 \\ 2 & -3 & 2 \end{vmatrix}}{\left|-4\hat{i}-8\hat{j}-8\hat{k}\right|}\right| = \left|\frac{(-12)-10(8)+2(-8)}{12}\right|$$

$= 9$ units

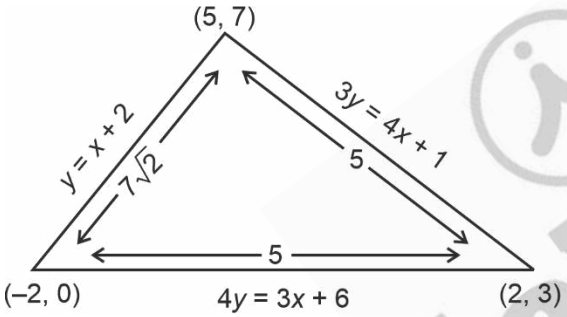
65. Let $y = x + 2$, $4y = 3x + 6$ and $3y = 4x + 1$ be three tangent lines to the circle $(x - h)^2 + (y - k)^2 = r^2$. Then $h + k$ is equal to

(1) $5\left(1+\sqrt{2}\right)$ (2) 5

(3) 6 (4) $5\sqrt{2}$

Answer (2)

Sol.



$$(h,k) = \left(\frac{5.5+5(-2)+14\sqrt{2}}{10+7\sqrt{2}}, \frac{35+21\sqrt{2}}{10+7\sqrt{2}}\right)$$

$$h+k = \frac{50+35\sqrt{2}}{10+7\sqrt{2}} = 5$$

66. Among the statements :

$(S1)$ $((p \vee q) \Rightarrow r) \Leftrightarrow (p \Rightarrow r)$

$(S2)$ $((p \vee q) \Rightarrow r) \Leftrightarrow ((p \Rightarrow r) \vee (q \Rightarrow r))$

(1) only $(S1)$ is a tautology

(2) only $(S2)$ is a tautology

(3) neither $(S1)$ nor $(S2)$ is a tautology

(4) both $(S1)$ and $(S2)$ are tautologies

Answer (3)

Sol. $S_1 : ((p \vee q) \Rightarrow r) \Leftrightarrow (p \Rightarrow r)$

$S_2 : ((p \vee q) \Rightarrow r) \Leftrightarrow ((p \Rightarrow r) \vee (q \Rightarrow r))$

In S_1 : If p = F, $q = T$, r = F then S_1 is false

In S_2 : if $P = T$, $q = F$, $r = F$ then S_2 is false

$\therefore$ Neither $S1$ nor $S2$ is a tautology

67. The coefficient of x^{301} in $(1+x)^{500} + x(1+x)^{499} + x^2(1+x)^{498} + \ldots\ldots + x^{500}$ is

(1) ${}^{501}C_{302}$ (2) ${}^{501}C_{200}$

(3) ${}^{500}C_{300}$ (4) ${}^{500}C_{301}$

Answer (2)

Sol. ${}^{500}C_{301} + {}^{499}C_{300} + {}^{498}C_{299} + \ldots + {}^{199}C_0$

$= {}^{500}C_{199} + {}^{499}C_{199} + {}^{498}C_{199} + \ldots + {}^{199}C_{89}$

$= {}^{501}C_{200}$

68. The minimum number of elements that must be added to the relation $R = \{a, b), (b, c)\}$ on the set $\{a, b, c\}$ so that it becomes symmetric and transitive is

(1) 7 (2) 3

(3) 5 (4) 4

Answer (1)

Sol. For symmetric $(b, a), (c, b) \in R$

For transitive $(a, c) \in R$

$\Rightarrow (c, a) \in R$

$\therefore (a, b), (b, a) \in R$

$\Rightarrow (a, a) \in R$

$(b, c), (c, b) \in R$

$\Rightarrow (b, b) \in R, (c, c) \in R$

7 elements must be added

69. If $P(h, k)$ be a point on the parabola $x = 4y^2$, which is nearest to the point $Q(0,33)$, then the distance of P from the directrix of the parabola $y^2 = 4(x + y)$ is equal to

(1) 4 (2) 6

(3) 8 (4) 2

Answer (2)

Sol. Equation of normal

$$y = -tx + 2 \cdot \frac{1}{16}t + \frac{1}{16}t^3$$

$$33 = \frac{t}{8} + \frac{t^3}{16}$$

$\Rightarrow t^3 + 2t = 528$

$t = 8$

$(at^2, 2at) = (4, 1)$

Distance from $x = -2$

70. If [t] denotes the greatest integer $\le t$, then the value of $\frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+\left[x^3\right]}dx$ is

(1) $e^7 - 1$ (2) $e^8 - 1$

(3) $e^9 - e$ (4) $e^8 - e$

Answer (4)

Sol. $I = \frac{3(e-1)}{e}\int_1^2 x^2 e^{[x]+\left[x^3\right]}dx$

$= \frac{3(e-1)}{e}\int_1^2 x^2 e^{1+\left[x^3\right]}dx \qquad (\because [x] = 1 \text{ when } x \in (12))$

$= 3(e-1)\int_1^2 x^2 e^{\left[x^3\right]}dx$

Let $x^3 = t$

$I = (e-1)\int_1^8 e^{[t]}dt$

$= \left(e^{-1}\right)\left(e^1 + e^2 + e^3 + ... + e^7\right)$

$= (e-1)e\frac{\left(e^7-1\right)}{e-1}$

$= e^8 - e$

71. Let a unit vector $\widehat{OP}$ makes angles α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$. If $\widehat{OP}$ is perpendicular to the plane through points (1, 2, 3), (2, 3, 4) and (1, 5, 7), then which one of the following is true?

(1) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(2) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(3) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(4) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

Answer (3)

Sol. Let $A \equiv (1, 2, 3)$, $B \equiv (2, 3, 4)$, $C \equiv (1, 5, 7)$

$\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 0 & 3 & 4 \end{vmatrix}$

$= \hat{i} - 4\hat{j} + 3\hat{k}$

$\widehat{OP} = \frac{\pm\left(\hat{i} - 4\hat{j} + 3\hat{k}\right)}{\sqrt{26}}$

Since $\cos\beta > 0$, take – sign

$\widehat{OP} = \frac{\hat{i} - 4\hat{j} + 3\hat{k}}{\sqrt{26}}$

$\Rightarrow \cos\alpha < 0, \cos\gamma < 0$

$\alpha, \gamma \in \left(\frac{\pi}{2}, \pi\right)$

72. Suppose $f : \mathbb{R} \to (0, \infty)$ be a differentiable function such that $5f(x+y) = f(x)\cdot f(y), \forall x, y \in \mathbb{R}$. If $f(3) = 320$, then $\sum_{n=0}^{5} f(n)$ is equal to

(1) 6825 (2) 6525

(3) 6875 (4) 6575

Answer (1)

Sol. $5f(x + y) = f(x)\cdot f(y)$

$5f(3) = f(1)\cdot f(2)$

$5f(2) = (f(1))^2$ $\qquad f(10) = 5$

$f(1) = 20$

$\Rightarrow f(1)\cdot\frac{(f(1))^2}{5} = 1600$

$\sum_{n=0}^{5} f(n) = f(0) + 20 + 80 + 320 + 1280 + 5120$

$= 1750 + 5120$

$= 6825$

73. Let $A = \begin{pmatrix} m & n \\ p & q \end{pmatrix}$, $d = |A| \neq 0$ and $|A - d(Adj\ A)| = 0$. Then

(1) $(1+d)^2 = m^2 + q^2$ (2) $(1+d)^2 = (m+q)^2$

(3) $1 + d^2 = (m+q)^2$ (4) $1 + d^2 = m^2 + q^2$

Answer (2)

Sol. $\left|A - d\begin{pmatrix} q & -n \\ -p & m \end{pmatrix}\right| = 0$

$\begin{vmatrix} m-qd & n(1+d) \\ p(1+d) & q-md \end{vmatrix} = 0$

$(m - qd)(q - md) = np(1 + d)^2$

$mq - (q^2 + m^2)d + qmd^2 = np(1 + d^2) + 2npd$

$d^2(mq - np) + 1(mq - np) = (2np + m^2 + q^2)d$

$(d^2 + 1)(mq - np) = (2np + m + a)d$

$d^2 + 1 = 2np + m^2 + q^2$

$2d = 2mq - 2np$

$\Rightarrow \quad (1 + d)^2 = (m + q)^2$

74. If $\tan15° + \frac{1}{\tan75°} + \frac{1}{\tan105°} + \tan195° = 2a$, then the value of $\left(a + \frac{1}{a}\right)$ is

(1) 4 (2) 2

(3) $4 - 2\sqrt{3}$ (4) $5 - \frac{3}{2}\sqrt{3}$

Answer (1)

Sol. $\tan15º + \tan15º - \tan15º + \tan15º$

$= 2\tan15º$

$= 2\left(2 - \sqrt{3}\right) = 2a \quad \Rightarrow a = 2 - \sqrt{3}$

$\therefore \frac{1}{a} + a \Rightarrow \left(2 + \sqrt{3}\right) + \left(2 - \sqrt{3}\right) = 4$

75. Let the system of linear equations

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

have infinitely many solutions. Then the system

$(k + 1)x + 2(k - 1)y = 7$

$(2k + 1)x + (k + 5)y = 10$

has :

(1) Unique solution satisfying x + y = 1

(2) Unique solution satisfying x – y = 1

(3) Infinitely many solutions

(4) No solution

Answer (1)

Sol. $x + y + kz = 2 \quad \ldots(1)$

$2x + 3y - z = 1 \quad \ldots(2)$

$3x + 4y + 2z = k \quad \ldots(3)$

(1) + (2)

$3x + 4y + z(k - 1) = 3$

Comparing with (3)

$k = 3$

Now, $4x + 5y = 7 \quad \Rightarrow 3x + 3y = 3$

$7x + 8y = 10$

as $\frac{4}{7} \neq \frac{5}{8}$

$\therefore$ unique solution satisfying $x + y = 1$

76. If the solution of the equation $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1$, $x \in \left(0, \frac{\pi}{2}\right)$, is $\sin^{-1}\left(\frac{\alpha + \sqrt{\beta}}{2}\right)$, where α, β are integers, then $\alpha + \beta$ is equal to

(1) 6

(2) 5

(3) 4

(4) 3

Answer (3)

Sol. $\log_{\cos x}\cot x + 4\log_{\sin x}\tan x = 1$

$\Rightarrow \log_{\cos x}\cot x - 4\log_{\sin x}\cot x = 1$

$\Rightarrow 1 - \log_{\cos x}\sin x - 4 - 4\log_{\sin x}\cos x = 1$

Let $\log_{\cos x}\sin x = t$

$t + \frac{4}{t} = 4$

$\Rightarrow t = 2$

$\sin x = \cos^2 x$

$\Rightarrow \sin x = 1 - \sin^2 x$

$\Rightarrow \sin^2 x + \sin x - 1 = 0$

$\Rightarrow \sin x = \frac{-1 \pm \sqrt{5}}{2}$

as $x \in \left(0, \frac{\pi}{2}\right)$

$\sin x = \frac{\sqrt{5} - 1}{2}$

$x = \sin^{-1}\left(\frac{-1 + \sqrt{5}}{2}\right)$

$\Rightarrow \alpha = -1, \beta = 5$

$\alpha + \beta = 4$

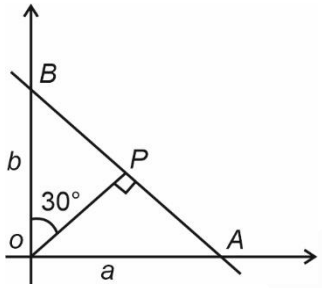
77. A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive direction of x-axis and y-axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y-axis and the area of ΔOAB is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to

(1) 196 (2) $\frac{196}{3}$

(3) $\frac{392}{3}$ (4) 98

Answer (3)

Sol. $\frac{1}{2}ab = \frac{98\sqrt{3}}{3}$



$\Rightarrow \sqrt{3}ab = 196$ ...(i)

$OP = OB\cos30° = OA\cos60°$

$\Rightarrow \frac{b\sqrt{3}}{2} = \frac{a}{2}$

$\Rightarrow \sqrt{3}b = a$ ...(ii)

By (i) and (ii)

$a^2 = 196$

$a = 14$

$b^2 = \frac{a^2}{3}$

$a^2 - b^2 = \frac{2a^2}{3} = \frac{392}{3}$

78. If $a_n = \frac{-2}{4n^2 - 16n + 5}$, then $a_1 + a_2 + \ldots\ldots + a_{25}$ is equal to

(1) $\frac{49}{138}$ (2) $\frac{52}{147}$

(3) $\frac{51}{144}$ (4) $\frac{50}{141}$

Answer (4)

Sol. $\sum_{i=1}^{25} a_i = \sum \frac{-2}{4n^2 - 16n + 15} = \sum \frac{-2}{(2n-5)(2n-3)}$

$= \sum_{i=1}^{25}\left(\frac{1}{2n-3} - \frac{1}{2n-5}\right)$

$= \left[\left(\frac{1}{-1} - \frac{1}{-3}\right) + \left(\frac{1}{1} - \frac{1}{-1}\right) + \left(\frac{1}{3} - \frac{1}{1}\right)\ldots\ldots\right.$

$= \frac{1}{2(25)-3} + \frac{1}{3} = \frac{50}{141}$

79. Let the solution curve $y = y(x)$ of the differential equation

$$\frac{dy}{dx} - \frac{3x^5\tan^{-1}\left(x^3\right)}{\left(1+x^6\right)^{3/2}}y = 2x\ \exp\left\{\frac{x^3 - \tan^{-1}x^3}{\sqrt{\left(1+x^6\right)}}\right\}$$

pass through the origin. Then $y(1)$ is equal to

(1) $\exp\left(\frac{1-\pi}{4\sqrt{2}}\right)$ (2) $\exp\left(\frac{4-\pi}{4\sqrt{2}}\right)$

(3) $\exp\left(\frac{4+\pi}{4\sqrt{2}}\right)$ (4) $\exp\left(\frac{\pi-4}{4\sqrt{2}}\right)$

Answer (2)

Sol. $\frac{dy}{dx} - \frac{3x^5\tan^{-1}\left(x^3\right)}{\left(1+x^6\right)^{\frac{3}{2}}}y = 2x\exp\left\{\frac{x^3 - \tan^{-1}x^3}{\sqrt{1+x^6}}\right\}$

$IF = e^{-\int\frac{3x^5\tan^{-1}\left(x^3\right)}{\left(1+x^6\right)^{\frac{3}{2}}}dx}$

Let $\tan^{-1}x^3 = t \Rightarrow \frac{3x^2}{1+x^6}dx = dt$

$\Rightarrow IF = e^{-\int\frac{\tan t}{\sec t}\cdot t\,dt} = e^{-\int\sin t\cdot t dt} = e^{t\cos t - \sin t}$

$\Rightarrow IF = e^{\frac{\tan^{-1}\left(x^3\right)}{\sqrt{1+x^6}} - \frac{x^3}{\sqrt{1+x^6}}}$

$\therefore$ Solution is

$$y\cdot e^{\frac{\tan^{-1}x^3}{\sqrt{1+x^6}} - \frac{x^3}{\sqrt{1+x^6}}} = \int 2x\,dx + c$$

$$\Rightarrow y\cdot e^{\frac{\tan^{-1}x^3 - x^3}{\sqrt{1+x^6}}} = x^2 + c$$

$y(0) = 0 \Rightarrow c = 0$

$x = 1$

$$y \cdot e^{\frac{\frac{\pi}{4}-1}{\sqrt{2}}} = 1$$

$$\Rightarrow \quad y = e^{\frac{1-\frac{\pi}{4}}{\sqrt{2}}}$$

$$\Rightarrow \quad y = e^{\frac{4-\pi}{4\sqrt{2}}}$$

80. If an unbiased die, marked with –2, –1, 0, 1, 2, 3 on its faces, is thrown five times, then the probability that product of the outcomes is positive is:

(1) $\frac{881}{2592}$

(2) $\frac{440}{2592}$

(3) $\frac{27}{288}$

(4) $\frac{521}{2592}$

Answer (4)

Sol. ${}^5C_0 \times 3^5 = 243$

${}^5C_2 \times 2^2 \times 3^3 = 1080$

${}^5C_4 \times 2^4 \cdot 3 = 240$

$\therefore$ required probability

$$= \frac{243 + 1080 + 240}{6 \times 6 \times 6 \times 6 \times 6} = \frac{521}{2592}$$

SECTION - B

Numerical Value Type Questions: This section contains 10 questions. In Section B, attempt any five questions out of 10. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the second decimal place; e.g. 06.25, 07.00, –00.33, –00.30, 30.27, –27.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.

81. Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one functions $f : S \to P(S)$, where $P(S)$ denote the power set of S, such that $f(n) \subset f(m)$ where $n < m$ is _____.

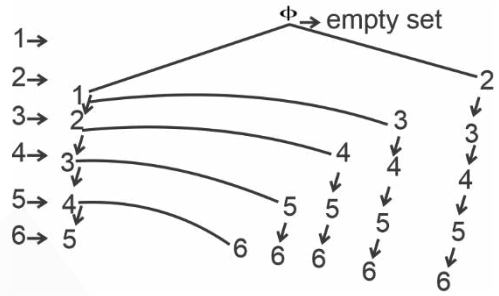
Answer (3240)

Sol. $\because \quad S = \{1, 2, 3, 4, 5, 6\}$

and $P(S) = \{\phi, \{1\}, \{2\}, \ldots, \{1, 2, 3, 4, 5, 6\}\}$

$f(n)$ corresponding a set having m elements which belongs to $P(S)$, should be a subset of $f(n + 1)$, so $f(n + 1)$ should be a subset of $P(S)$ having at least $m + 1$ elements.

Now if $f(1)$ has one element then $f(2)$ has 3, $f(3)$ has 3 and so on and $f(6)$ has 6 elements. Total number of possible functions = 6! = 720 …(1) if $f(1)$ has no elements (*i.e.* null set ϕ) then



Each index number represents the number of elements in respective rows

Taking every series of arrow and counting number of such possible functions (sets)

$$= {}^6C_2 \times {}^4C_1 \times {}^3C_1 \times {}^2C_1 + {}^6C_1 \times {}^5C_2 \times {}^3C_1 \times {}^2C_1$$
$$+ {}^6C_1 \times {}^5C_1 \times {}^4C_2 \times {}^2C_1 + {}^6C_1 \times {}^5C_1 \times {}^4C_1 \times {}^3C_2$$
$$+ {}^6C_1 \times {}^5C_1 \times {}^4C_1 \times {}^3C_1 \times {}^2C_2 + {}^6C_1 \times {}^5C_1 \times {}^4C_1 \times {}^3C_1 \times {}^2C_1$$
$$= 2520 \qquad \ldots(2)$$

From (1) and (2) : Total number of functions

= 2520 + 720

= 3240

82. Number of 4-digit numbers (the repeation of digits is allowed) which are made using the digits 1, 2, 3 and 5, and are divisible by 15, is equal to _____

Answer (21)

Sol. Digits 1, 2, 3, 5 and number should be divisible by 15 (i.e., divisible by both 3 and 5)

So,

…(5)

Case-I: $5 \to 1, 2 \to 1, 1 \to 2 = \frac{3!}{2!} = 3$

$5 \to 1, 3 \to 1, 2 \to 2 = \frac{3!}{2!} = 3$

$5 \to 1, 3 \to 2, 1 \to 1 = \frac{3!}{2!} = 3$

Case-II: $5 \to 2, 3 \to 1, 2 \to 1 = 3! = 6$

$5 \to 2, 1 \to 2 = \frac{3!}{2!} = 3$

Case-III: $5 \to 3, 3 \to 1 = \frac{3!}{2!} = 3$

$\therefore$ Total no. = 21

83. If $\lambda_1 < \lambda_2$ are two values of λ such that the angle between the planes $P_1 : \vec{r}\left(3\hat{i} - 5\hat{j} + \hat{k}\right) = 7$ and $P_2 : \vec{r}\left(\lambda\hat{i} + \hat{j} - 3\hat{k}\right) = 9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is _____.

Answer (315)

Sol. $P_1 : \vec{r} \cdot \left(3\hat{i} - 5\hat{j} + \hat{k}\right) = 7$

$P_2 : \vec{r} \cdot \left(\lambda\hat{i} + \hat{j} - 3\hat{k}\right) = 9$

Let angle between P_1 and P_2 is θ

Then $\cos\theta = \frac{3\lambda - 5 - 3}{\sqrt{35}\sqrt{\lambda^2 + 10}}$

But $\sin\theta = \frac{2\sqrt{6}}{5}$

$\therefore \quad \frac{(3\lambda - 8)^2}{35\left(\lambda^2 + 10\right)} = 1 - \frac{24}{25}$

$\Rightarrow 5(9\lambda^2 + 64 - 48\lambda) = 7\lambda^2 + 70$

$\Rightarrow 38\lambda^2 - 240\lambda + 250 = 0$

$\Rightarrow 19\lambda^2 - 120\lambda + 125 = 0$

$\Rightarrow (19\lambda - 25)(\lambda - 5) = 0$

$\therefore \quad \lambda_1 = \frac{25}{19}, \lambda_2 = 5$

So, point (50, 50, 2)

$\therefore \quad d = \frac{|150 - 250 + 2 - 7|}{\sqrt{35}} = 315$

84. Let $\sum_{n=0}^{\infty} \frac{n^3((2n)!) + (2n-1)(n!)}{(n!)((2n)!)} = ae + \frac{b}{e} + c$, where $a, b, c \in \mathbb{Z}$ and $e = \sum_{n=0}^{\infty} \frac{1}{n!}$ Then $a^2 - b + c$ is equal to _____.

Answer (26)

Sol. $\sum_{n=0}^{\infty} \frac{n^3(2n!) + (2n-1)(n!)}{n! \cdot (2n)!}$

$= \sum_{n=0}^{\infty} \frac{n^3}{n!} + \frac{2n-1}{2n!}$

$= \sum_{n=0}^{\infty} \frac{3}{(n-2)!} + \frac{1}{(n-3)!} + \frac{1}{(n-1)!} + \frac{1}{(2n-1)!} - \frac{1}{(2n)!}$

$= 3e + e + e - \frac{1}{e}$

$= 5e - \frac{1}{e}$

$\therefore \quad a = 5, b = -1, c = 0$

$\therefore \quad a^2 - b + c = 26$

85. Let $z = 1 + i$ and $z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$. Then $\frac{12}{\pi}\arg(z_1)$ is equal to _____.

Answer (09)

Sol. $z = 1 + i$

$z_1 = \frac{1 + i\bar{z}}{\bar{z}(1 - z) + \frac{1}{z}}$

$= \frac{z(1 + i\bar{z})}{|z|^2 (1 - z) + 1}$

$= \frac{(1 + i)(1 + i(1 - i))}{2(1 - 1 - i) + 1}$

$z_1 = 1 - i$

$\arg z_1 = \tan^{-1}\left(\frac{-1}{1}\right) = \frac{3\pi}{4}$

$\frac{12}{\pi}\arg(z_1) = \frac{3\pi}{4} \cdot \frac{12}{\pi}$

$= 9$

86. Let $f^1(x) = \frac{3x + 2}{2x + 3}$, $x \in R - \left\{\frac{-3}{2}\right\}$

For $n \geq 2$, define $f^n(x) = f^1 \, of \, ^{n-1}(x)$.

If $f^5(x) = \frac{ax + b}{bx + a}$, gcd(a, b) = 1, then $a + b$ is equal to _____.

Answer (3125)

Sol. $f'(x) = \frac{3x + 2}{2x + 3} \, x \in R - \left\{-\frac{3}{2}\right\}$

$f^5(x) = f_o \, f_o \, f_o \, f_o \, f(x)$

$f_o f(x) = \frac{13x + 12}{12x + 13}$

$f_o \, f_o \, f_o \, f_o \, f(x) = \frac{1563x + 1562}{1562x + 1563}$

$\equiv \frac{ax + b}{bx + a}$

$\therefore \quad a = 1563, b = 1562$

$= 3125$

87. $\lim\limits_{x\to 0}\dfrac{48}{x^4}\int\limits_0^x \dfrac{t^3}{t^6+1}dt$ is equal to _____.

Answer (12)

Sol. $\lim\limits_{x\to 0}\dfrac{48}{x^4}\int\limits_0^x \dfrac{t^3}{t^6+1}dt.$

$\lim\limits_{x\to 0}\dfrac{48}{4x^3}\cdot\left(\dfrac{x^3}{x^6+1}\right)$

$\lim\limits_{x\to 0}\dfrac{12}{x^6+1}=12$

88. The mean and variance of 7 observations are 8 and 16 respectively. If one observation 14 is omitted and a and b are respectively mean and variance of remaining 6 observation, then $a+3b-5$ is equal to _____.

Answer (37)

Sol. $\sum x_i = 7\times 8 = 56$

$\dfrac{\sum x_i^2}{n}-\left(\dfrac{\sum x_i}{n}\right)^2=16$

$\dfrac{\sum x_i^2}{7}-64=16$

$\sum x_i^2 = 560$

When 14 is omitted

$\sum x_i = 56-14=42$

New mean = $a=\dfrac{\sum x_i}{6}=7$

$\sum x_i^2 = 560-196=364$

new variance, $b=\dfrac{\sum x_i^2}{6}-\left(\dfrac{\sum x_i}{6}\right)^2$

$=\dfrac{364}{6}-49=\dfrac{35}{3}$

$3b = 35$

$a+3b-5 = 7+35-5 = 37$

89. If the equation of the plane passing through the point 1, 1, 2) and perpendicular to the line $x-3y+2z-1=0=4x-y+z$ is $Ax+By+Cz=1$, then $140(C-B+A)$ is equal to _____.

Answer (15)

Sol. Line of intersection of the planes $x-3y+2z-1=0$ and $4x-y+z=0$ is normal $(\vec{n})$ to the required plane.

$$\vec{n}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -3 & 2\\ 4 & -1 & 1\end{vmatrix}=-\hat{i}+7\hat{j}+11\hat{k}$$

Equation of plane is

$-x+7y+11z=\lambda$

It passes through (1, 1, 2)

$\therefore\quad \lambda = 28$

So, the plane is

$-x+7y+11z=28$

$\Rightarrow\quad \dfrac{-1}{28}x+\dfrac{7}{28}y+\dfrac{11}{28}z=1$

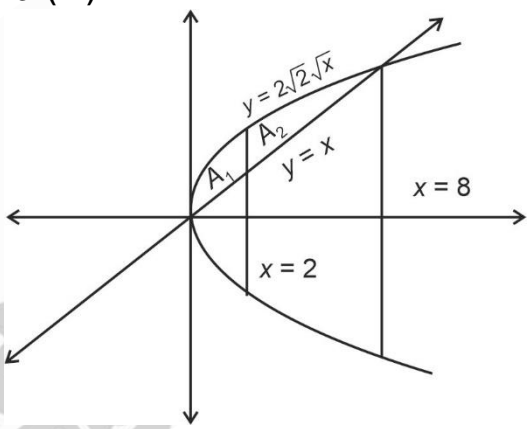
$A=\dfrac{-1}{28}, B=\dfrac{7}{28}, C=\dfrac{11}{28}$

$140(C-B+A) = 15$

90. Let α be the area of the larger region bounded by the curve $y^2=8x$ and the line $y=x$ and $x=2$, which lies in the first quadrant. Then the value of 3α is equal to _____.

Answer (22)

Sol.



$A_1=\int\limits_0^2 2\sqrt{2}\sqrt{x}-x\,dx$

$=2\sqrt{2}\times\dfrac{2}{3}\cdot x^{\frac{3}{2}}-\dfrac{x^2}{2}\Big]_0^2$

$=\dfrac{4\sqrt{2}}{3}\times 2\sqrt{2}-2$

$=\dfrac{16}{3}-2=\dfrac{10}{3}$

$A_2=\int\limits_2^8 2\sqrt{2}\sqrt{x}-x\,dx=\dfrac{4\sqrt{2}}{3}\cdot x^{\frac{3}{2}}-\dfrac{x^2}{2}\Big]_2^8$

$=\dfrac{4\sqrt{2}}{3}\left(16\sqrt{2}-2\sqrt{2}\right)-30$

$=\dfrac{112}{3}-30=\dfrac{22}{3}$

$A_2 > A_1 \Rightarrow 3\alpha = 22$

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